REAL Neural Substrate:
Emergent Computation and Cross-Task Transfer Through Local Allostatic Learning in a Multi-Agent Routing Network

[Author names omitted for review]
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Abstract

We present **REAL Neural Substrate**, a multi-agent architecture in which useful computation, cross-task transfer, and structural adaptation emerge from purely local allostatic dynamics — without backpropagation, a global loss function, or any centralised planner. Each node is an independent agent operating the *Recognize–Evaluate–Act–Learn* (REAL) loop subject to a metabolic energy budget (ATP). Durable learning is encoded in a *memory substrate* that lowers the cost of successful routing behaviors across sessions. We evaluate the system on two benchmarks: (1) a family of context-dependent routing tasks (CVT-1) requiring the network to infer hidden task structure from local feedback alone, and (2) room occupancy prediction from real-world time-series sensor data. On CVT-1, the substrate achieves meaningful task performance within a single 18-packet session — approximately 8–9× more sample-efficient than an online-trained Elman RNN under the same data budget. Sequential $A \to B \to C$ transfer chains demonstrate that substrate carryover compounds without catastrophic forgetting. On the occupancy dataset, a fully deployed V3 harness achieves a warm F1 score of **0.952**, compared to 0.963 for a fully-supervised MLP trained on 2.7× more data — the first result in this line of work that is competitive with a conventional supervised baseline on a real-world task. The remaining gap is attributable to a recall deficit on the minority class, a structural consequence of class-uniform feedback weighting rather than any fundamental limitation of the architecture. These results establish REAL Neural Substrate as a credible, empirically grounded platform for low-data, continually-adapting, substrate-based computation.

1 Introduction

Contemporary deep learning achieves remarkable performance by optimising billions of passive parameters against a global objective over large, curated datasets [LeCun et al., 2015]. This paradigm has proven powerful but carries structural costs: enormous data requirements, expensive retraining when task conditions change, and a learning signal that must be broadcast simultaneously across the entire model. These costs are especially limiting in settings where data arrives incrementally, tasks shift over time, or hardware constraints preclude centralised computation.

A complementary tradition — drawing on Hebbian associativity [Hebb, 1949], allostatic regulation [Sterling & Laughlin, 2012], and local structural plasticity [Bhatt et al., 2009] — suggests that useful computation can arise from networks of locally-acting agents responding only to their immediate metabolic state, without any global teacher. Project REAL operationalises this intuition.

Our approach. We construct a graph of node-agents. Each node executes an independent perception–action–consolidation loop, spends a finite ATP budget on routing actions, and earns replenishment when its actions contribute to correctly delivering a packet to the network sink. Long-run regularities are encoded in a *maintained substrate* (M_s): a persistent cost-reduction structure that biases future action selection toward historically rewarded behaviors. Crucially, this substrate persists across sessions, enabling *carryover*: the routing expertise accumulated during task A provides a structural head start on task B.

Contributions.

1. A formal description of the REAL engine and its six-primitive coherence model, including the two-layer episodic/substrate memory system (§3).
2. A multi-agent substrate instantiation (Phase 8) with context-indexed action support, metabolic admission control, and topology morphogenesis (§4).
3. Experimental evidence of positive $A \to B$ and $A \to B \to C$ transfer without catastrophic forgetting, sample efficiency advantages over online-trained neural baselines, and difficulty-correlated self-regulating morphogenesis (§5).
4. The first real-world evaluation of the substrate on a time-series classification task (occupancy prediction), reaching F1 = 0.952 — within 0.011 of a fully-supervised MLP (§7).

2 Related Work

Local learning rules. Hebbian plasticity [Hebb, 1949] and spike-timing-dependent plasticity (STDP) [Bi & Poo, 1998] demonstrate that local co-activity rules can produce useful representations without global error signals. REAL extends this lineage by pairing Hebbian-style consolidation with an explicit metabolic budget and a persistence mechanism that survives session boundaries.

Reinforcement learning. RL agents [Sutton & Barto, 2018] optimise a scalar reward signal, typically with global value-function updates. REAL nodes have *no external reward*: they receive ATP as a byproduct of contributing to correct routing, and their "objective" is metabolic survival. The system is closer in spirit to intrinsically motivated exploration [Oudeyer & Kaplan, 2007] than to standard RL.

Continual and meta-learning. Continual learning research addresses catastrophic forgetting through methods such as EWC [Kirkpatrick et al., 2017] and progressive networks [Rusu et al., 2016]. Meta-learning approaches [Finn et al., 2017] seek rapid adaptation from few examples via learned inductive biases. REAL addresses both goals simultaneously through substrate persistence: prior session structure lowers action costs for related tasks without requiring an outer optimisation loop or explicit memory replay.

Multi-agent emergent communication. Work on emergent communication in multi-agent systems [Lazaridou & Baroni, 2020] explores how agents can develop shared protocols through local interaction. REAL nodes develop routing specialisation through the same mechanism — differential survival — rather than through a designed communication protocol.

Algorithm 1: REAL Node Cycle

1. $o \leftarrow \textsc{Observe}(\mathcal{O})$ *(local state only)*
2. $\hat{a} \leftarrow \Psi(o,\, M_s)$ *(substrate-biased selection)*
3. ATP $-= c(\hat{a})$ *(metabolic spend)*
4. $\textsc{Execute}(\hat{a})$
5. $r \leftarrow \textsc{AwaitFeedback}()$ *(sequential upstream signal)*
6. ATP $+= r$ *(ATP replenishment)*
7. $e \leftarrow \langle o,\, \hat{a},\, r,\, \Phi(o, \hat{a}, r)\rangle$
8. $H_e \leftarrow H_e \cup \{e\}$ *(episodic trace)*
9. $\Gamma(H_e,\, H_c,\, M_s)$ *(promote if attractor pattern)*

Neuromorphic and biologically-inspired computing. Spiking neural networks [Maass, 1997] and neuromorphic hardware [Davies et al., 2018] implement local plasticity at scale. REAL is architecturally adjacent: the ATP budget is a functional analogue of membrane potential, and the substrate is an analogue of long-term potentiation. The current implementation is software-only; neuromorphic deployment is a natural future direction.

3 Architecture

3.1 The REAL Engine

The REAL engine formalises each node agent as a tuple

$$\langle \mathcal{S},\, \mathcal{A},\, c,\, \mathcal{O},\, \Phi,\, H,\, \Psi,\, \Gamma,\, \Omega \rangle$$

where $\mathcal{S}$ is the local state space, $\mathcal{A}$ the action vocabulary, $c : \mathcal{A} \to \mathbb{R}_{\geq 0}$ the ATP cost function, $\mathcal{O}$ the observable neighbourhood, Φ the coherence scoring model, $H = \langle H_e, H_c \rangle$ the two-layer memory, Ψ the selection policy, Γ the consolidation pipeline, and Ω the substrate protocol.

Each cycle executes Algorithm **??**.

3.2 The Six Coherence Primitives

Coherence scoring Φ decomposes into six dimensions derived from the E^2 relational framework:

1. **Continuity** — does the action preserve the node's historical routing function?

2. **Vitality** — is the node actively participating in useful routing (receiving ATP)?
3. **Contextual fit** — does the selected action match the context-indexed support in M_s?
4. **Differentiation** — does the node offer routing value distinct from its neighbours?
5. **Accountability** — is the causal flow of the action traceable through the substrate?
6. **Reflexivity** — does the node's self-model in M_s reflect its actual performance history?

3.3 Two-Layer Memory

Memory is structured as a fast episodic trace H_e and a slow consolidated layer H_c, with a maintained substrate M_s that physically reduces action costs. Formally, for an action a in context k:

$$c_t(a \mid k) \;=\; c_0(a) \cdot \big(1 - \alpha \cdot M_s(a,k)\big)$$

where $c_0(a)$ is the baseline cost, $\alpha \in (0,1)$ is the substrate discount factor, and $M_s(a,k) \in [0,1]$ is the accumulated support for action a under context k. Promotion from H_e to M_s is gated by a rolling performance threshold, preventing the substrate from encoding transient noise.

4 Phase 8: Multi-Agent Substrate

4.1 Routing Environment

Phase 8 instantiates N node-agents on a directed graph. A source node injects packets; a sink node returns sequential upstream feedback proportional to packet delivery quality. Each packet carries a payload (a 4-bit string), an optional context bit, and a path history. The correct transformation of the payload (e.g., ROTATE_LEFT_1 or XOR_MASK_1010) is context-dependent and unknown to individual nodes — it must be inferred from the statistics of which actions earn ATP.

Node actions include: ROUTE(v), REST, INHIBIT(v), INVEST(e), and TRANSFORM-ROUTE(T, v). The substrate maintains two support structures:

$$\text{Edge support:} \quad M_s^e(u \to v) \in [0,1] \tag{1}$$

$$\text{Action support:} \quad M_s^a(a,k) \in [0,1] \tag{2}$$

Edge support is task-agnostic; action support is context-indexed. This distinction is consequential for transfer (see §**??**).

4.2 Admission Control

A source admission gate prevents overload by restricting ingress when the node's ATP balance falls below a threshold. This enforces metabolic discipline: the substrate cannot accumulate spurious patterns from a flooded routing graph.

4.3 Morphogenesis

When a node's ATP surplus exceeds a threshold τ_{atp} and routing feedback falls below a threshold τ_{fb}, it can bud a new edge or spawn a child node. New nodes receive a grace period during which upkeep costs are waived, giving them time to find productive routing roles before their metabolic obligations begin. A context-resolution gate prevents morphogenesis from firing before the node has accumulated sufficient context confidence.

5 Experimental Design

5.1 CVT-1: Context-Dependent Routing Tasks

The core synthetic benchmark is the CVT-1 task family. A session consists of P packets; each packet's payload must be transformed according to the active context bit $k \in \{0,1\}$. Three tasks define a structured transfer landscape:

Task	Context 0	Context 1
A	ROTATE_LEFT_1	XOR_1010
B	ROTATE_LEFT_1	XOR_0101
C	XOR_1010	XOR_0101

Tasks A and B share context 0; B and C share context 1; A and C share a transform (A's context-1 equals C's context-0). This structure makes transfer directional and asymmetric. The small-topology benchmark uses 6 nodes and $P = 18$ packets; the large-topology benchmark uses 10 nodes and $P = 36$ packets with 3-way source branching.

5.2 Neural Baselines

Three online-trained baselines use the same 18-example CVT-1 sequence under a predict-then-update regime: **MLP-explicit** (5 inputs, context provided), **MLP-latent** (4 inputs, context withheld), and **RNN-latent** (Elman RNN, hidden-state sequence memory). All baselines use stochastic gradient descent with a rolling window criterion: $\geq 85\%$ exact-match accuracy over the most recent 6 predictions.

Table 1: Small-topology ($N = 6$, $P = 18$) carryover results. Exact: mean exact matches / 18. BitAcc: mean bit accuracy.

Condition	Exact	BitAcc
Cold Task B	3.9	0.477
Warm substrate-only $A \to B$	7.3	0.586
Warm full $A \to B$	**10.6**	**0.704**
Cold Task A (latent)	2.2	0.461
Warm $A \to B$ (visible)	6.2	0.506
Warm $A \to B$ (latent)	**7.0**	**0.517**

5.3 Occupancy Prediction

The occupancy benchmark uses `occupancy_synth_v1.csv`: 1,344 rows of five environmental sensor readings (CO_2, temperature, humidity, light, motion) labelled occupied / unoccupied. The temporal 70/30 split preserves causal ordering. A frozen MLP trained by standard supervised gradient descent on the full training split serves as the supervised baseline.

The REAL harness evolved through three versions:

V1 Continuous train/eval; feedback suppressed during eval; no carryover test; no context bit. *(Invalid — see §7.1)*

V2 Feedback at full fraction during eval; `fresh_eval` carryover protocol; CO_2-derived binary context bit.

V3 Online running context (4-code, multi-sensor); multihop routing topology; admission-source ingress; explicit `fresh_session_eval` and `persistent_eval` protocols; multi-seed sweep with auto CPU budgeting.

6 Results I: CVT-1 Routing Tasks

6.1 Baseline Transfer

Table 1 shows the core small-topology carryover results. Warm full carryover ($A \to B$) achieves 10.6 exact matches out of 18, compared to 3.9 for cold-start Task B — a 2.7× improvement. Substrate-only carryover (7.3 exact) confirms that the structural substrate, independent of episodic replay, is the primary carrier of transfer benefit.

Table 2: Sequential transfer on Task C ($N = 6$, $P = 18$).

Condition	Exact / 18	BitAcc
Cold C	4.6	0.433
$B \to C$	6.0	0.456
$A \to C$	7.6	0.528
$A \to B \to C$	**7.6**	**0.561**

6.2 Latent Context: Transfer-Safe by Construction

When context labels are withheld during training (*latent mode*), nodes must infer the active context from feedback statistics. After tuning the commitment streak from 3 to 2 observations (threshold tightened from 0.75 to 0.78), Task B latent cold-start rose from 4.0 to 8.6 exact matches. More strikingly, latent $A \to B$ transfer (7.0 exact) now exceeds visible $A \to B$ transfer (6.2 exact).

The mechanism is architectural. Visible training writes context-specific action supports $M_s^a(a, k = 1)$ keyed to task A's context-1 transform. When task B redefines context 1, these supports become *context poison* — an inhibitory prior that costs approximately 3–4 exact matches. Latent training writes context-agnostic supports $M_s^a(a, k = \varnothing)$, which carry no binding to any specific label. They cannot poison and tend to benefit by providing routing scaffold.

6.3 Sequential Transfer Without Catastrophic Forgetting

Table 2 reports $A \to B \to C$ transfer results. Both the three-step chain and a direct $A \to C$ skip achieve 7.6 exact matches on Task C, but through complementary mechanisms:

$A \to B \to C$ B training deeply reinforces $M_s^a(\text{XOR_0101}, k = 1)$ — the correct context-1 action for C. Massive context-1 lift (+0.463 bit accuracy) partially offset by A+B context-0 poison (−0.14). Net: highest bit accuracy (0.561).

$A \to C$ **direct** A's $M_s^a(\text{XOR_1010}, k = 1)$ coincidentally equals C's context-0 transform. Positive context-0 prior; weaker context-1 lift. Net: lower bit accuracy (0.528).

Critically, B training does not erase A's value — it compounds it for the contexts where A and C align while adding new context-1 value. This is graceful specialisation, not catastrophic overwrite.

Table 3: Sample efficiency comparison on CVT-1 ($P = 18$). Criterion: $\geq 85\%$ exact-match over last 6 predictions.

Model	Examples to criterion	Ratio vs REAL
MLP-explicit	∼54	3.0×
MLP-latent	>360 (fails)	—
RNN-latent	∼144–162	8–9×
REAL	**18**	1×

Table 4: Morphogenesis results across topologies ($P = 18$ small, $P = 36$ large). Delta: growth minus fixed (exact matches).

Topology	Condition	Δ Exact	Earned%	Win%
Small ($N = 6$)	Task B cold	+0.0	20	20
Small ($N = 6$)	$A \to B$ warm	+1.2	80	60
Large ($N = 10$)	Task A cold	−1.6	100	40
Large ($N = 10$)	Task B cold	**+7.4**	100	**80**
Large ($N = 10$)	Task C cold	−1.0	80	40
Large ($N = 10$)	$A \to B$ warm	+2.2	100	80

6.4 Neural Baseline Comparison

Table 3 compares REAL against online-trained neural baselines under the rolling-window criterion. The RNN-latent baseline — the fairest analogue to REAL's hidden-context setting — requires approximately 144–162 examples to reach criterion. REAL reaches comparable milestone behavior in a single 18-packet session: approximately 8–9× more sample-efficient. The MLP-latent baseline cannot solve the bimodal mapping at all within 20 epochs.

6.5 Morphogenesis

Morphogenesis is evaluated by the *earned growth rate* (fraction of seeds where newly budded nodes receive non-zero routing feedback) and *win rate* (fraction where morphogenesis improves over fixed topology).

Table 4 reveals a consistent pattern: morphogenesis provides benefit proportional to the gap between current performance and the topology maximum. On the small ($N = 6$) topology, cold-start earned growth is only 20%; on the large ($N = 10$) topology, it reaches 100% for all task-carrying scenarios. This is not a configuration artifact — it reflects the ATP surplus gate operating correctly. High-performing tasks exhaust ATP efficiently, leaving no surplus to trigger budding. Struggling tasks accumulate surplus from failed routing attempts, triggering growth precisely when it is needed.

Table 5: V1 vs. V2 occupancy evaluation results (seed 13). DR: packet delivery ratio.

Configuration	Eval Acc	Eval F1	DR
V1 (invalid)	0.772	0.032	∼0.32
V2 live	0.892	0.748	0.846
V2 carryover	0.900	0.757	0.831
V2 live + CO_2	**0.900**	**0.766**	**0.977**
V2 carry + CO_2	0.896	0.764	0.970

7 Results II: Occupancy Prediction

7.1 V1 Evaluation Failure

The initial occupancy harness produced an eval F1 of 0.032, which appeared to indicate architectural failure. Post-hoc analysis identified three compounding design flaws, each sufficient to render the evaluation invalid:

1. **ATP starvation.** Feedback was suppressed during evaluation (`training=False`), cutting the ATP replenishment signal. Mean dropped packets surged from 0.96 (training) to 17.35 (eval).
2. **No carryover test.** The evaluation never isolated substrate transfer as an independent variable. The core value proposition was never measured.
3. **Dormant context layer.** No context bit was injected, leaving the context-indexed action support layer $M_s^a(a, k)$ unwritten throughout.

7.2 V2: Structural Fixes and Initial Results

V2 addressed each flaw with a targeted change: (1) `eval_feedback_fraction` = 1.0, (2) `fresh_eval` carryover protocol, (3) CO_2-derived binary context bit. Table 5 reports the outcome. Enabling feedback alone raised eval F1 from 0.032 to 0.748 — a 23× improvement attributable entirely to removing a structural defect, not to any model change. The best V2 configuration achieved F1 = 0.766 with a delivery ratio of 0.977, using 500 training episodes (37% of available data).

7.3 V3: Architectural Alignment and Full Results

V3 replaced V2's heuristic workarounds with REAL-native mechanisms: (a) *online running context* —

Table 6: V3 run progression (2026-03-20, seed 13 unless noted). All use `fresh_session_eval`. ER: efficiency ratio (warm/cold delivery).

Run	Sess	W-Acc	W-F1	C-F1	ER
Smoke (10)	10	0.955	0.727	0.750	0.991
Mid-scale (50)	50	0.965	0.918	0.922	0.989
Full (62)	62	**0.981**	**0.952**	0.919	**0.998**
Sweep avg (3 seeds)	62	**0.973**	—	—	0.992

Table 7: Comparison of MLP baseline vs. V3 REAL (seed 13, `fresh_session_eval`). Training data volume differs: MLP saw ∼1,344 examples; V3 REAL trained on 62 sessions.

System	Acc	Prec	Rec	F1
MLP (full data)	**0.985**	**0.976**	**0.951**	**0.963**
V3 warm (fresh)	0.981	0.988	0.920	0.952
V3 cold (fresh)	0.969	1.000	0.851	0.919
F1 gap				*−0.011*

a 4-code context derived continuously from multi-sensor readings, replacing the single CO_2 bit; (b) *multihop routing topology*, engaging the full substrate machinery; (c) *admission-source ingress*, restoring metabolic gating.

Table 6 shows the progression of V3 runs on 2026-03-20. As the harness matured and training sessions increased, F1 rose monotonically. The full 62-session run achieves warm F1 = 0.9524. A 3-seed sweep (seeds 13, 23, 37) confirms stability: mean warm accuracy 0.9734, mean efficiency ratio 0.9915.

7.4 Comparison with MLP Baseline

Table 7 summarises the full V3 vs. MLP comparison. V3 warm F1 of 0.952 lies within 0.011 of the MLP's 0.963. The gap is almost entirely a recall difference: MLP recall 0.951 vs. V3 warm recall 0.920. Both systems achieve near-identical precision (∼0.98).

The MLP is optimising a classification loss directly. REAL is routing packets and inferring occupancy indirectly through context-indexed substrate activity, without any explicit class objective. The remaining recall gap is attributable to the absence of class-weighted feedback weighting: REAL does not currently provide stronger ATP signals for the minority (occupied) class, so it is less sensitive to edge cases in that class. This is a correctable algorithmic choice, not a fundamental architectural limit.

Notably, V3 cold-start recall (0.851) is substantially lower than warm (0.920), confirming that substrate carryover contributes materially to occupancy-class sensitivity. The warm substrate, loaded from the trained session, improves recall for the minority class by approximately 7 percentage points relative to a cold start.

8 Discussion

8.1 Carryover as the Primary Mechanism

Across both CVT-1 and occupancy benchmarks, substrate carryover is the primary driver of performance improvements. The distinction between full episodic and substrate-only carryover (Table 1) confirms that the structural cost-reduction layer — not runtime episodic replay — carries the transfer benefit. The occupancy fresh-session protocol isolates this cleanly: a fresh system loaded with the training substrate outperforms a cold-start system without ever having been in the same session as the eval data.

8.2 Context Binding: Power and Poison

Context-indexed action supports are the substrate's primary specialisation mechanism. They enable rapid context-conditional routing when the context geometry is stable across tasks. But they become poison when the task changes the meaning of a context label — the prior binding then inhibits rather than assists.

The latent path avoids this by construction: supports indexed to $k = \varnothing$ carry no context binding and therefore no poison. The practical implication is that latent carryover provides a reliable positive transfer floor across arbitrary task geometry, while visible carryover is maximally helpful when geometry is aligned and maximally harmful when it is not. Choosing between them requires knowing (or inferring) the degree of overlap between source and target task context mappings.

8.3 Morphogenesis as a Headroom Detector

The consistent finding that morphogenesis helps where performance is low and disrupts where performance is already high is not a limitation — it is the ATP surplus gate functioning as a *headroom detector*. This is a useful emergent property: the growth system is effectively asking "do I have routing capacity to spare?" and answering "no" in cases where the

fixed topology is already efficient. The practical implication is that morphogenesis configurations should be calibrated per scenario or per performance tier, rather than set globally.

8.4 The Occupancy Gap: Recall, Not Architecture

The remaining 0.011 F1 gap between V3 warm and the MLP is a recall gap attributable to uniform feedback weighting. The occupancy dataset is approximately 80% unoccupied, and REAL provides the same ATP signal per correctly routed packet regardless of class. The MLP's loss function implicitly upweights the minority class through standard gradient magnitude dynamics. Adding class-weighted feedback to REAL is a one-parameter change (`feedback_amount` by label) and is the most targeted next step.

8.5 Limitations

Several limitations constrain the current findings:

- The small-topology transfer results and March 17 latent-transfer results come from different benchmark slices and should not be combined into a single table without noting this.
- All occupancy V3 runs flagged `not_applicable_all_eval_contexts_seen`: the temporal split did not produce held-out context codes. True latent transfer in the occupancy domain remains untested.
- Morphogenesis under admission-managed load is blocked by a known observation gap: `queue_pressure` stays near zero because admission control prevents overflow, making it indistinguishable from low-load to the growth gate.
- The neural baseline comparison is approximate for single-pass exact match counts; the epoch-scan comparison is the more reliable claim.

9 Conclusion

We have presented REAL Neural Substrate, a multi-agent architecture in which useful computation and cross-task transfer emerge from local allostatic dynamics without global optimisation. Experiments on the CVT-1 routing benchmark establish that substrate carryover is real and measurable, that latent context inference is transfer-safe by architectural construction, that morphogenesis self-regulates toward structural headroom, and that the system is approximately 8–9× more sample-efficient than an online-trained RNN on the same hidden-context task. Sequential $A \to B \to C$ transfer chains demonstrate graceful specialisation rather than catastrophic forgetting.

On room occupancy prediction — the first real-world dataset applied to this architecture — a fully deployed V3 harness achieves warm F1 of 0.952, within 0.011 of a fully-supervised MLP trained on 2.7× more data. The remaining gap is a class-recall asymmetry correctable by a single hyperparameter change.

Together, these results position REAL Neural Substrate as a credible experimental platform for low-data, continually-adapting, substrate-based computation and suggest that the bottom-up, metabolically-constrained paradigm deserves serious attention alongside the dominant top-down, gradient-based paradigm.

Acknowledgments

[To be completed by authors.]

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A CVT-1 Configuration × Task Family Performance Map

Table 8 summarises the prescription emerging from the March 18 configuration sweep across task families A, B, and C.

Table 8: Configuration prescription by task family and conditions.

Task characteristics	Best cold config	Best transfer config
Short horizon, unambiguous context (A1–A2)	fixed-visible	growth-visible + substrate-only
Large scale, unambiguous context (A3–A4)	fixed-visible	growth-latent + full episodic
Hidden memory (B2–B4)	fixed-visible	growth-latent + full episodic
Partial ambiguity (C1–C2)	fixed-visible	fixed-visible (→ task b)
Deep ambiguity, short topo. (C3)	fixed-latent	fixed-latent
Deep ambiguity, large topo. (C4)	fixed-latent	growth-latent (→ task b)
Multi-hop cyclic sequences	fixed-latent	latent cyclic

B V3 Occupancy Harness Configuration

The default V3 harness configuration for the full benchmark run is:

```
context_mode:          online_running_context
topology_mode:         multihop_routing
ingress_mode:          admission_source
eval_mode:             fresh_session_eval
eval_feedback_fraction: 1.0
feedback_amount:       0.18
packet_ttl:            8
window_size:           5
train_session_fraction: 0.70
selector_seed:         13
```

The 3-seed sweep used `--selector-seeds 13 23 37` with `auto_cpu_target_fraction = 0.75`, yielding a worker budget of 15 on the test machine (3 concurrent seeds × 5 eval workers per seed).